

Algorithm 1: Prompt → Geometry → Route

Parse the Constraints

Alg. 1, L1

"Write an assistant

response that is

comprehensive

and

concise."

C1

C2

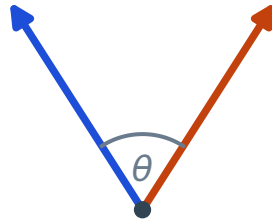
Embed the Geometry

Alg. 1, L2

$$\hat{\rho}_{ij} = \cos(\Phi(C_i), \Phi(C_j))$$

$\Phi(\mathbf{C1})$

$\Phi(\mathbf{C2})$



$$\hat{\rho} = \cos \theta = 0.42$$

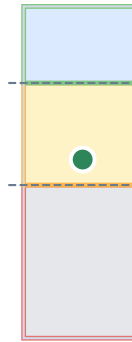
conflict proxy

Budget for Feasibility

Alg. 1, L3-4

$$\text{ratio} = \delta / \delta_{\min}$$

$$\frac{\delta}{\delta_{\min}} = 1.4$$



One-Shot

2

Staged

1.2

Decompose

Selectively Route

Alg. 1, L6-12



One-Shot

$$\delta / \delta_{\min} > 2$$



Staged

$$\hat{\rho} < 0.5$$



Decompose

else

*before generation
< 10 ms*